

PROGRAM-1: Swap Two 8-bit Numbers

```
LDA 8500H
MOV B, A
LDA 8501H
STA 8500H
MOV A, B
STA 8501H
HLT
```

PROGRAM-2: Reverse 8-bit Number

```
LDA 8500H
RRC
RRC
RRC
RRC
STA 8501H
HLT
```

PROGRAM-3: Even / Odd → Display DD / FF

```
LDA 8000H
ANI 01H
JZ EVEN
MVI A, FFH
STA 8050H
HLT
EVEN: MVI A, DDH
STA 8050H
HLT
```

PROGRAM-4: Swap Two 16-bit Numbers

```
LHLD 8000H
XCHG
LHLD 8002H
SHLD 8000H
XCHG
SHLD 8002H
HLT
```

PROGRAM-5: 4-Digit BCD Addition

```
LHLD 8500H
XCHG
LHLD 8502H
MOV A, L
ADD E
DAA
STA 8504H
MOV A, H
ADC D
DAA
STA 8505H
HLT
```

PROGRAM-6: Largest of Two Numbers

```
LXI H, 8000H
MOV B, M
INX H
MOV A, M
CMP B
JNC STORE
MOV A, B
STORE: STA 8050H
HLT
```

PROGRAM-7 (8051): Smallest of Two Numbers

```
LXI H, 8000H
MOV B, M
INX H
MOV A, M
CMP B
JC STORE
MOV A, B
STORE: STA 8050H
HLT
```

PROGRAM-8: Sum of Digits of 8-bit Number

```
LDA 8050H
MOV B, A
ANI 0FH
```

```
MOV C, A
MOV A, B
ANI F0H
RRC
RRC
RRC
RRC
ADD C
STA 8050H
HLT
```

PROGRAM–9: Fibonacci Series (8085)

```
LXI H, 8050H
XRA A
MOV B, A
MOV M, A
INR A
INX H
MOV M, A
MVI C, 08H
LOOP: ADD B
MOV B, A
INX H
MOV M, A
DCR C
JNZ LOOP
HLT
```

PROGRAM–10: Division of Two 8-bit Numbers

```
LXI H, 4450H
MOV B, M
MVI C, 00H
INX H
MOV A, M
NEXT: CMP B
JC LOOP
SUB B
INR C
JMP NEXT
LOOP: STA 4452H
MOV A, C
STA 4453H
```

HLT

11. 8085 Program to Calculate the Sum of First Ten Natural Numbers

```
LXI H, 8000H
MOV C, M
XRA A
MOV B, A
MOV A, M
LOOP: ADD C
JNC SKIP
INR B
SKIP: DCR C
JNZ LOOP
LXI H, 8050H
MOV M, A
INX H
MOV M, B
HLT
```

12. 8085 Program to Count Number of 1's in an 8-bit Number

```
LDA 8000H
MOV H, A
MVI D, 00H
MVI C, 08H
LOOP: RRC
JNC SKIP
INR D
SKIP: DCR C
JNZ LOOP
MOV A, D
STA 8050H
HLT
```